

## Exogenous surfactant treatments for neonatal respiratory distress syndrome and their potential role in the adult respiratory distress syndrome.

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Exogenous surfactant therapy has been recognised as an approach to alleviating the surfactant-deficiency state for 3 decades. Natural and lipid-extracted surfactants derived from amniotic fluid, lung lavage, or lung homogenates are being used in worldwide clinical trials in premature infants. These studies are demonstrating a generally favourable influence on lung function by improving oxygenation and reducing the risk for pneumothorax and pulmonary interstitial emphysema. In some studies, reduction in death and the occurrence of bronchopulmonary dysplasia have been found. Numerous questions are unresolved and pharmacokinetic data are limited in preterm infants. Artificial surfactants are similarly under evaluation but current data demonstrate less overall effect. Adult respiratory distress syndrome has also been treated with exogenous surfactants. Although complex in terms of multiple initiating factors and in terms of high permeability of surfactant inhibitors, further studies are under way to determine the ideal methods of administration to enhance distribution and to monitor surfactant function *in vivo*.

## Liver RBFOX2 regulates cholesterol homeostasis via Scarb1 alternative splicing in mice.

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RNA alternative splicing (AS) expands the regulatory potential of eukaryotic genomes. The mechanisms regulating liver-specific AS profiles and their contribution to liver function are poorly understood. Here, we identify a key role for the splicing factor RNA-binding Fox protein 2 (RBFOX2) in maintaining cholesterol homeostasis in a lipogenic environment in the liver. Using enhanced individual-nucleotide-resolution ultra-violet cross-linking and immunoprecipitation, we identify physiologically relevant targets of RBFOX2 in mouse liver, including the scavenger receptor class B type I (Scarb1). RBFOX2 function is decreased in the liver in diet-induced obesity, causing a Scarb1 isoform switch and alteration of hepatocyte lipid homeostasis. Our findings demonstrate that specific AS programmes actively maintain liver physiology, and underlie the lipotoxic effects of obesogenic diets when dysregulated. Splice-switching oligonucleotides targeting this network alleviate obesity-induced inflammation in the liver and promote an anti-atherogenic lipoprotein profile in the blood, underscoring the potential of isoform-specific RNA therapeutics for treating metabolism-associated diseases.

## A single-cell atlas of glioblastoma evolution under therapy reveals cell-intrinsic and cell-extrinsic therapeutic targets.

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Recent longitudinal studies of glioblastoma (GBM) have demonstrated a lack of apparent selection pressure for specific DNA mutations in recurrent disease. Single-cell lineage tracing has shown that GBM cells possess a high degree of plasticity. Together this suggests that phenotype switching, as opposed to genetic evolution, may be the escape mechanism that explains the failure of precision therapies to date. We profiled 86 primary-recurrent patient-matched paired GBM specimens with single-nucleus RNA, single-cell open-chromatin, DNA and spatial transcriptomic/proteomic assays. We found that recurrent

GBMs are characterized by a shift to a mesenchymal phenotype. We show that the mesenchymal state is mediated by activator protein 1. Increased T-cell abundance at recurrence was prognostic and correlated with hypermutation status. We identified tumor-supportive networks of paracrine and autocrine signals between GBM cells, nonmalignant neuroglia and immune cells. We present cell-intrinsic and cell-extrinsic targets and a single-cell multiomics atlas of GBM under therapy.

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## Protein Analysis of Glioblastoma Primary and Posttreatment Pairs Suggests a Mesenchymal Shift at Recurrence.

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Glioblastomas (GBM) are aggressive brain tumors that inevitably recur despite surgical resection, chemotherapy, and radiation. The degree to which recurrent GBM retains its initial immunophenotype is incompletely understood. We generated tissue microarrays of paired initial and posttreatment GBM (3 pairs positive and 17 negative for IDH1R132H) from the same patients and made comparisons in the IDH1R132H-negative group for immunohistochemical and gene expression differences between primary and recurrent tumors. In initial tumors, immunopositivity for Ki-67 in > 20% of tumor cells was associated with shorter progression-free and overall survival. Recurrent tumors showed decreased staining for CD34 suggesting lower vessel density. A subset of tumors showed increased staining for markers associated with the mesenchymal gene expression pattern, including CD44, phosphorylated STAT3, and YKL40. Recurrent tumors with the greatest increase in mesenchymal marker expression had rapid clinical progression, but no difference in overall survival after second surgery. Comparison of protein and gene expression data from the same samples revealed a poor correlation. A subset of tumors (15%) showed loss of neurofibromin protein in both initial and recurrent tumors. These data support the notion that GBM progression is associated with a shift toward a mesenchymal phenotype in a subset of tumors and this may portend a more aggressive behavior.

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## Proneural-Mesenchymal Transition: Phenotypic Plasticity to Acquire Multitherapy Resistance in Glioblastoma.

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Glioblastoma (GBM) is an extremely aggressive tumor of the central nervous system, with a prognosis of 12-15 months and just 3-5% of survival over 5 years. This is mainly because most patients suffer recurrence after treatment that currently consists in maximal resection followed by radio- and chemotherapy with temozolomide. The recurrent tumor shows a more aggressive behavior due to a phenotypic shift toward the mesenchymal subtype. Proneural-mesenchymal transition (PMT) may represent for GBM the equivalent of epithelial-mesenchymal transition associated with other aggressive cancers. In this review we frame this process in the high degree of phenotypic inter- and intra-tumor heterogeneity of GBM, which exists in different subtypes, each one characterized by further phenotypic variability in its stem-cell compartment. Under the selective pressure of different treatment agents PMT is induced. The mechanisms involved, as well as the significance of such event in the acquisition of a multitherapy resistance phenotype, are taken in consideration for future perspectives in new anti-GBM therapeutic options.

## VC-resist glioblastoma cell state: vessel co-option as a key driver of chemoradiation resistance.

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Glioblastoma (GBM) is a highly lethal type of cancer. GBM recurrence following chemoradiation is typically attributed to the regrowth of invasive and resistant cells. Therefore, there is a pressing need to gain a deeper understanding of the mechanisms underlying GBM resistance to chemoradiation and its ability to infiltrate. Using a combination of transcriptomic, proteomic, and phosphoproteomic analyses, longitudinal imaging, organotypic cultures, functional assays, animal studies, and clinical data analyses, we demonstrate that chemoradiation and brain vasculature induce cell transition to a functional state named VC-Resist (vessel co-opting and resistant cell state). This cell state is midway along the transcriptomic axis between proneural and mesenchymal GBM cells and is closer to the AC/MES1-like state. VC-Resist GBM cells are highly vessel co-opting, allowing significant infiltration into the surrounding brain tissue and homing to the perivascular niche, which in turn induces even more VC-Resist transition. The molecular and functional characteristics of this FGFR1-YAP1-dependent GBM cell state, including resistance to DNA damage, enrichment in the G2M phase, and induction of senescence/stemness pathways, contribute to its enhanced resistance to chemoradiation. These findings demonstrate how vessel co-option, perivascular niche, and GBM cell plasticity jointly drive resistance to therapy during GBM recurrence.

## Gene regulatory network topology governs resistance and treatment escape in glioma stem-like cells.

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Poor prognosis and drug resistance in glioblastoma (GBM) can result from cellular heterogeneity and treatment-induced shifts in phenotypic states of tumor cells, including dedifferentiation into glioma stem-like cells (GSCs). This rare tumorigenic cell subpopulation resists temozolomide, undergoes proneural-to-mesenchymal transition (PMT) to evade therapy, and drives recurrence. Through inference of transcriptional regulatory networks (TRNs) of patient-derived GSCs (PD-GSCs) at single-cell resolution, we demonstrate how the topology of transcription factor interaction networks drives distinct trajectories of cell state transitions in PD-GSCs resistant or susceptible to cytotoxic drug treatment. By experimentally testing predictions based on TRN simulations, we show that drug treatment drives surviving PD-GSCs along a trajectory of intermediate states, exposing vulnerability to potentiated killing by siRNA or a second drug targeting treatment-induced transcriptional programs governing non-genetic cell plasticity. Our findings demonstrate an approach to uncover TRN topology and use it to rationally predict combinatorial treatments that disrupts acquired resistance in GBM.

## Phenotype Switching in Glioblastoma Promotes Therapy Escape.

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A shift to a mesenchymal phenotype characterizes recurrence in glioblastoma and is mediated by AP1.

## Soluble ICAM-1 a Pivotal Communicator between Tumors and Macrophages, Promotes Mesenchymal Shift of Glioblastoma.

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Despite aggressive clinical treatment, recurrence of glioblastoma multiforme (GBM) is unavoidable, and the clinical outcome is still poor. A convincing explanation is the phenotypic transition of GBM cells upon aggressive treatment such as radiotherapy. However, the microenvironmental factors contributing to GBM recurrence after treatment remain unexplored. Here, it is shown that radiation-treated GBM cells produce soluble intercellular adhesion molecule-1 (sICAM-1) which stimulates the infiltration of macrophages, consequently enriching the tumor microenvironment with inflammatory macrophages. Acting as a paracrine factor, tumor-derived sICAM-1 induces macrophages to secrete wingless-type MMTV integration site family, member 3A (WNT3A), which promotes a mesenchymal shift of GBM cells. In addition, blockade of either sICAM-1 or WNT3A diminishes the harmful effect of radiation on tumor progression. Collectively, the findings indicate that cellular crosstalk between GBM and macrophage through sICAM-1-WNT3A oncogenic route is involved in the mesenchymal shift of GBM cells after radiation, and suggest that radiotherapy combined with sICAM-1 targeted inhibition would improve the clinical outcome of GBM patients.

## Transcriptional regulatory networks of tumor-associated macrophages that drive malignancy in mesenchymal glioblastoma.

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Glioblastoma (GBM) is a complex disease with extensive molecular and transcriptional heterogeneity. GBM can be subcategorized into four distinct subtypes; tumors that shift towards the mesenchymal phenotype upon recurrence are generally associated with treatment resistance, unfavorable prognosis, and the infiltration of pro-tumorigenic macrophages. We explore the transcriptional regulatory networks of mesenchymal-associated tumor-associated macrophages (MA-TAMs), which drive the malignant phenotypic state of GBM, and identify macrophage receptor with collagenous structure (MARCO) as the most highly differentially expressed gene. MARCO<sup>high</sup> TAMs induce a phenotypic shift towards mesenchymal cellular state of glioma stem cells, promoting both invasive and proliferative activities, as well as therapeutic resistance to irradiation. MARCO<sup>high</sup> TAMs also significantly accelerate tumor engraftment and growth in vivo. Moreover, both MA-TAM master regulators and their target genes are significantly correlated with poor clinical outcomes and are often associated with genomic aberrations in neurofibromin 1 (NF1) and phosphoinositide 3-kinases/mammalian target of rapamycin/Akt pathway (PI3K-mTOR-AKT)-related genes. We further demonstrate the origination of MA-TAMs from peripheral blood, as well as their potential association with tumor-induced polarization states and immunosuppressive environments. Collectively, our study characterizes the global transcriptional profile of TAMs driving mesenchymal GBM pathogenesis, providing potential therapeutic targets for improving the effectiveness of GBM immunotherapy.